EEE 4308: Open-ended Lab

In addition to the group project, each group must complete an open-ended lab assignment. The task is to design and implement a digital logic-based Tea/Coffee Vending Machine system.

Open-ended Lab Tasks:

Open-ended lab will be performed in two phases –

- Software phase
- Hardware phase

Software Phase:

Step 1:

Design a digital logic circuit for an automatic Tea/Coffee Vending Machine. The system will take a 3-bit input to select the type of beverage. The output will be an 8-bit binary code. The first 3 bits of the output represent the beverage type, and the last 5 bits represent the Group Number (assigned by project supervisor).

Input–Output Specification:

Beverage Type	Input	Output
Raw Tea	000	100 + GroupNo(5-bit)
Tea + Milk	001	101 + GroupNo(5-bit)
Tea + Sugar	010	110 + GroupNo(5-bit)
Tea + Milk + Sugar	011	111 + GroupNo(5-bit)
Raw Coffee	100	000 + GroupNo(5-bit)
Coffee + Milk	101	001 + GroupNo(5-bit)
Coffee + Sugar	110	010 + GroupNo(5-bit)
Coffee + Milk + Sugar	111	011 + GroupNo(5-bit)

Additional System Requirements

- Add a 1-bit Enable input (E).
- When E = 1, the system produces the correct 8-bit output.
- When E = 0, the output must be 00000000 regardless of selection.

Step 2:

Implement the above-designed circuit using Proteus.

- You have to implement both schematic and PCB layout.
- For the PCB layout, design must be constrained within a 4-inch x 4-inch board area.

To learn how to design PCB in proteus, you should go through these tutorials:

- https://youtu.be/5TTkJ2CHMvQ
- https://youtu.be/TKhw6VZHwzU

Deadline for Software Phase:

Each of the groups have to submit their proteus files (in a zip file) as well as a software phase report (in PDF format) within the deadline. The report should contain –

i. Brief description of the problem statement.
ii. A comprehensive description of your design, including theoretical calculations and any simplifications made.
iii. Screenshots of proteus schematic and PCB layout
iv. Contributions of each member. One member from each group should submit the required documents in this google classroom.

Hardware Phase:

In this phase you will be taught how to fabricate PCBs by etching process, and each group will have to fabricate the PCB they have designed in software phase. Most of the necessary components will be provided by the lab for this fabrication process.